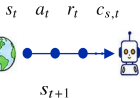


1

## Rollout



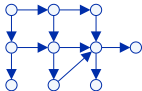
Collect transitions

 $(s_t, a_t, r_t, \{c_{k,t}\}, s_{t+1})$ 

2

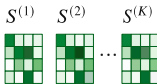
## Build flow inputs

Discrete state graph

empirical flow  $\hat{F}$ score matrix  $D$ capacities  $S^{(k)}$ 

3

## Compose capacities

 $S$  $S = \text{Compose}_{\varepsilon}(\{S^{(k)}\})$ 

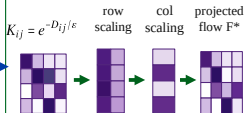
one matrix, not K duals

4

## Safe flow projection

$$F^* = \gamma\text{-Flow-Sinkhorn}(D, \hat{F}, S, \varepsilon)$$

Sinkhorn iterations



repeat to convergence

certificate:  $F^* \leq S$ 

5

## Weighted PPO update

weights

$$w_t = \text{clip}\left(\frac{F^*}{\max(\hat{F}, \delta)}\right)$$

PPO objective (clipped)

$$\mathcal{L}(\theta) = \mathbb{E}_t \left[ w_t \min \left( r_t \hat{A}_t, \text{clip}(r_t, 1 - \delta_{\text{clip}}, 1 + \delta_{\text{clip}}) \hat{A}_t \right) \right]$$

updated policy  $\pi_{\theta}$ 